

Freight Rate Prediction Using Machine Learning

Machine Learning Engineer Assessment

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1 Objective

The objective of this project is to predict the posted freight rate for a shipment using shipment, market, temporal, and geographic information. The labeled development dataset contains 48,000 rows and includes variables such as distance, equipment type, weight, market index, quote signal, pickup/delivery coordinates, and date. The final model is then used to generate predictions for the 12,000 unlabeled loads in `validation.csv`.

The main evaluation metrics used during development were Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), R^2 , and Mean Absolute Percentage Error (MAPE).

2 Data Preparation and Feature Engineering

Initial exploration identified missing values in `weight` and `market_index`, as well as a small number of invalid negative weights. Missing rows were removed from the labeled development data, negative weights were excluded, and duplicate rows were checked.

The following engineered features were added:

- day-of-week and month extracted from `date`;
- weekend and month-end indicators;
- sine/cosine encodings for day-of-week and month;
- geographic pickup and delivery regions using K-Means clustering with 10 clusters.

K-Means was fitted only on the training portion of the data to avoid information leakage. The same fitted clustering model was later reused for validation and final inference.

The main implementation is contained in the project notebook and is organized into sections for cleaning, feature engineering, model comparison, outlier investigation, model tuning, and final inference. The reusable prediction artifacts are saved as `freight_rate_model.pkl` and `geographic_kmeans.pkl`.

3 Validation Strategy

Because freight rates may change over time, the development data was sorted chronologically before splitting. A 70/15/15 split was used:

- 70% training set;
- 15% validation set;
- 15% test set.

The validation set was used for model comparison and hyperparameter selection. The test set was kept untouched until the final model had been selected. This produces a more realistic estimate of performance on future freight loads than a random split.

4 Model Comparison

Several regression algorithms were evaluated using the same feature set and validation procedure. These included Ridge Regression, Random Forest, HistGradientBoosting, an MLP neural network,

XGBoost, and CatBoost.

After outlier handling, the validation results were:

Table 1: Validation model comparison after outlier handling

Model	MAE	RMSE	R^2	MAPE
CatBoost	68.03	234.63	0.9692	4.44%
HistGradientBoosting	87.22	247.19	0.9658	5.17%
Ridge Regression	101.50	250.44	0.9649	7.54%
XGBoost	110.59	261.49	0.9618	7.15%
Random Forest	117.56	274.20	0.9580	6.94%
MLP Neural Network	126.70	266.84	0.9602	8.38%

CatBoost achieved the lowest validation MAE and MAPE and was therefore selected for tuning. The chosen configuration used 800 iterations, learning rate 0.03, depth 6, and `12_leaf_reg=3`.

5 Outlier Investigation

The initial model showed several very large residuals. Inspection showed that these loads had extremely high `posted_rate` values while the available input features did not show similarly extreme behavior. An IQR-based analysis identified an upper threshold of approximately \$6,450.35.

A total of 255 observations, approximately 0.54% of the cleaned development data, exceeded this threshold and were removed before retraining. This step substantially reduced RMSE and improved R^2 .

A limitation is that some removed observations may represent legitimate loads influenced by variables that are not present in the dataset. For this reason, the outlier handling is documented explicitly and no further target-based filtering was performed.

6 Final Model Performance

After model selection, the tuned CatBoost model was retrained using the combined training and validation data and evaluated once on the held-out test set.

Table 2: Final held-out test performance

Metric	Result
MAE	\$105.38
RMSE	\$248.34
R^2	0.9670
MAPE	6.41%

Permutation importance showed that `distance` was the dominant predictor, accounting for roughly 92.6% of normalized permutation importance. Equipment type was the second most important feature, followed by weight and the geographic-region features. This indicates that the model relies heavily on shipment distance, which is reasonable for freight-rate prediction but also represents a potential robustness limitation if market pricing behavior changes.

7 Final Validation Predictions

The final saved model was loaded and applied to every row in `validation.csv`. The same date feature engineering and the saved K-Means model were applied before prediction. Predictions were

rounded to two decimal places and written to:

```
validation_predictions.csv
```

with exactly the required columns:

```
load_id,predicted_rate
```

The inference code is implemented in the notebook/script section that loads `freight_rate_model.pkl`, loads `geographic_kmeans.pkl`, transforms `validation.csv`, and saves the completed prediction file.

8 Fixed December Prediction Chart

The provided `score.py` requires a second file containing 31 predictions for a fixed shipment from Lexington to Fort Wayne, 360 miles, Dry Van, 32,000 lb, with only the date changing from December 1 to December 31, 2025.

The model was used to fill the `predicted_rate` column in `december_predictions.csv`. The supplied scorer was then executed using:

```
python3 score.py \  
--predictions validation_predictions.csv \  
--december-predictions december_predictions.csv
```

The scorer validates both files and generates `scorer_results/candidate_december.png`. Insert the generated chart below before final submission.

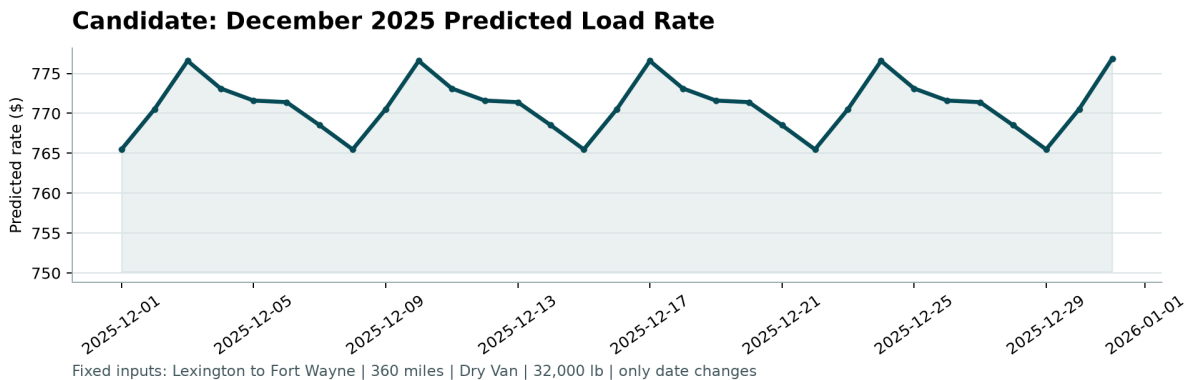


Figure 1: Fixed December 2025 predicted load-rate chart generated by `score.py`.

9 Conclusion

The project evaluated multiple regression approaches and selected CatBoost based on chronological validation performance. After documented outlier handling and hyperparameter tuning, the final held-out test performance reached an R^2 of 0.9670 with an MAE of approximately \$105. The final pipeline was then used to generate predictions for all 12,000 validation loads, together with the required fixed December prediction chart.

The main limitations are the strong dependence on distance, the possibility that some removed target outliers were legitimate, and the absence of potentially useful freight variables such as urgency, accessorial charges, and other operational conditions. Despite these limitations, the final pipeline is reproducible, uses an untouched test set for final evaluation, and produces the required submission outputs.